Day 26: Overdispersion and inference for Poisson regression

Recap: where we left off

We are modeling the number of anti-discrimination `laws` in each state using `percent_urban` and `historical` (`dem`, `gop`, or `swing`) with a **Poisson regression** model:

$$Y_i \sim \text{Poisson}(\lambda_i) \qquad \log(\lambda_i) = \beta_0 + \beta_1 \text{percent_urban}_i + \beta_2 \text{gop}_i + \beta_3 \text{swing}_i$$

```
full_glm <- glm(laws ~ percent_urban + historical, data = equality_index,
family = "poisson")
```

term	estimate	std. error	z	p-value
(Intercept)	0.7590	0.3269	2.32	0.020
percent_urban	0.0314	0.0037	8.42	<0.001
historicalgop	-1.6121	0.1332	-12.10	<0.001
historicalswing	-0.8288	0.1015	-8.17	<0.001

Example: In binomial logistic regression, what is the response variable? How is a Poisson count different?

Example: Interpret the `percent_urban` coefficient in context.

The Poisson model makes three assumptions:

1. **Linearity**: $\log(\lambda_i)$ is a linear function of the predictors
2. **Mean = Variance**:
3. **Independence**: the observations are independent

Today: what do we do when the second assumption fails?

Overdispersion

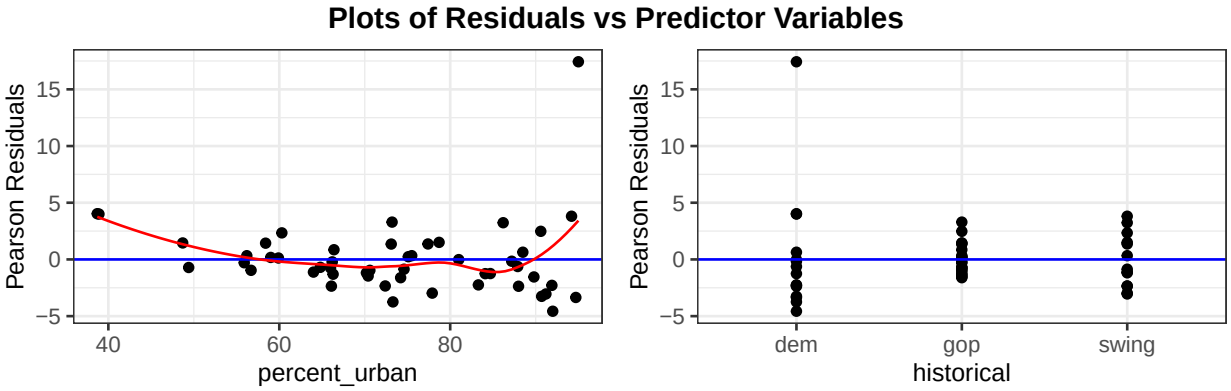
> **Overdispersion**
>
> **Overdispersion** means the data are *more* variable than a Poisson model allows: $Var(Y_i)$_____λ_i.

- Typical causes: counts that are clustered, differences between units that we haven't measured, events that are not independent.
- Consequence: the Poisson standard errors are too _____, so confidence intervals are too _____ and p-values are too _____.

Pearson residuals

> **Pearson residuals for a Poisson model**
>
> $$\text{Pres}_i = \frac{Y_i - \widehat{\text{mean}(Y_i)}}{\widehat{SD(Y_i)}} = _____$$
>
> - If the $\hat{\lambda}_i$ are big enough (about 5 or more), the Pearson residuals behave approximately like draws from a _____ distribution if the Poisson model is correct.
> - If the data are overdispersed, the standard deviation of the Pearson residuals will be _____

```
resid_xpanel(full_glm, type = "pearson", smoother = TRUE, nrow = 1)
```



What are we looking for in these plots?

Goodness-of-fit test

Just as in binomial logistic regression, we can test whether the model fits using the **Pearson χ^2 statistic**:

$$X^2 = \sum_{i=1}^{n} \frac{(Y_i - \hat{\lambda}_i)^2}{\hat{\lambda}_i} = \sum_{i=1}^{n} \text{Pres}_i^2$$

> **Goodness-of-fit test**
>
> **Hypotheses:** H_0: _____ vs. H_a: the model is inadequate
> **Test statistic:** X^2
> **Reference distribution:** χ^2 with $df =$ _____
> **p-value:** the area _____ X^2 (upper tail)

```
X2 <- sum(residuals(full_glm, type = "pearson")^2)
X2
```

```
[1] 516.1012
```

```
df.residual(full_glm)
```

```
[1] 46
```

To find the p-value:

```
1 - pchisq(____, ____)
```

Conclusion:

> **Are the means big enough?**
>
> The χ^2 approximation (for both the test and the Pearson residuals) needs the fitted means $\hat{\lambda}_i$ to be large enough (about 5 or more).

```
summary(fitted(full_glm))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  1.962   3.532   7.267  13.460  20.429  42.016 
```

Should we trust the test here? Why or why not?

Estimating the dispersion parameter

If X^2 is much bigger than $n-(p+1)$, the data are overdispersed. We can measure *how much* with the **dispersion parameter** ψ:

> **Quasipoisson model**
>
> $$Var(Y_i) = \psi\lambda_i \qquad \widehat{\psi} = \frac{X^2}{n-(p+1)}$$

Calculate $\widehat{\psi}$ for the `equality_index` model, and say what it means in context:

Fitting the quasipoisson model

```
full_quasi <- glm(laws ~ percent_urban + historical, data = equality_index,
                  family = "quasipoisson")
```

term	estimate	std. error	t	p-value
(Intercept)	0.7590	1.0948	0.69	0.492
percent_urban	0.0314	0.0125	2.51	0.015
historicalgop	-1.6121	0.4463	-3.61	<0.001
historicalswing	-0.8288	0.3399	-2.44	0.019

Compare with the Poisson fit:

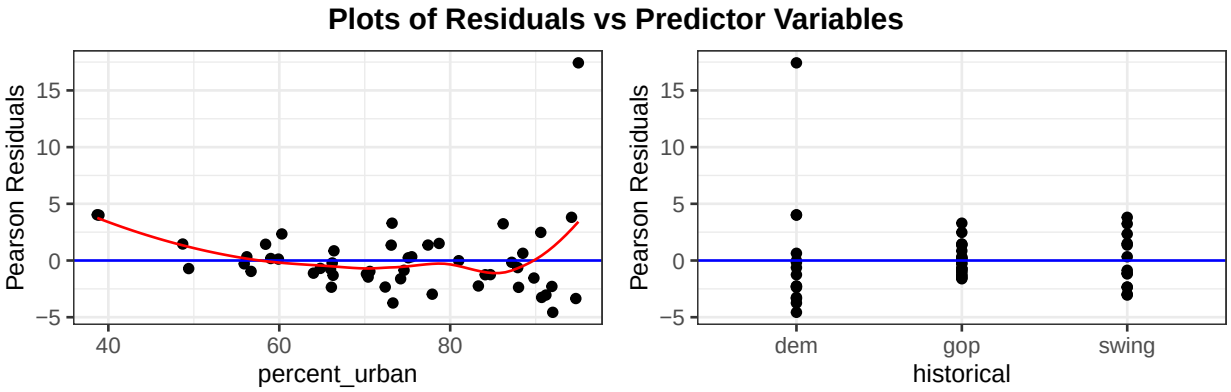
term	estimate	SE (Poisson)	SE (quasi)	ratio
(Intercept)	0.7590	0.3269	1.0948	3.3496
percent_urban	0.0314	0.0037	0.0125	3.3496
historicalgop	-1.6121	0.1332	0.4463	3.3496
historicalswing	-0.8288	0.1015	0.3399	3.3496

- The estimates _____ change.
- Every quasipoisson SE is the Poisson SE multiplied by _____ $= 3.35$.
- Test statistics are compared to a _____ distribution (with $n - (p + 1)$ df) instead of a $N(0, 1)$ distribution.

Residuals for the quasipoisson model

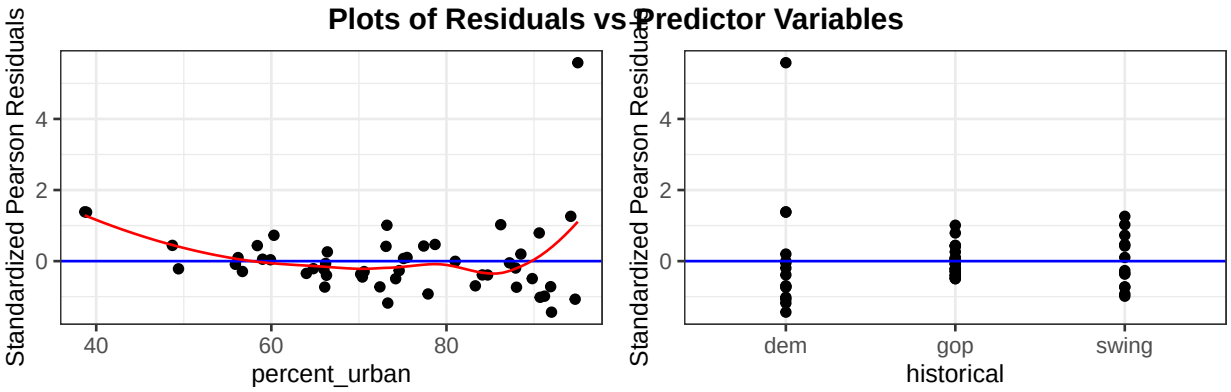
The ordinary Pearson residuals still use the Poisson standard deviation, so they still look overdispersed:

```
resid_xpanel(full_quasi, type = "pearson", smoother = TRUE, nrow = 1)
```



Standardized Pearson residuals use the quasipoisson standard deviation $SD(Y_i) = \sqrt{\psi \lambda_i}$ (and also adjust for leverage):

```
resid_xpanel(full_quasi, type = "stand.pearson", smoother = TRUE, nrow = 1)
```



Does the quasipoisson model fix the problem?

Inference for the coefficients

Confidence intervals and tests for β_j

	Poisson	Quasipoisson
Confidence interval	$\hat{\beta}_j \pm z^* SE(\hat{\beta}_j)$	$\hat{\beta}_j \pm t^*_{n-(p+1)} SE_{quasi}(\hat{\beta}_j)$
Test of $H_0: \beta_j = 0$	$z = \dfrac{\hat{\beta}_j}{SE}$, compared to $N(0,1)$	$t = \dfrac{\hat{\beta}_j}{SE_{quasi}}$, compared to $t_{n-(p+1)}$

We build intervals on the **log scale** first, then exponentiate. For a C-unit increase in x_j, multiply the estimate *and* the SE by C **before** exponentiating:

$$\left(e^{C\widehat{\beta}_j - z^* \cdot C \cdot SE},\ \ e^{C\widehat{\beta}_j + z^* \cdot C \cdot SE}\right)$$

Example: Calculate 95% confidence intervals for the effect of a **10 percentage point** increase in `percent_urban` under both models.

```
b <- coef(full_glm)[2]
se_pois <- summary(full_glm)$coefficients[2, 2]
se_quasi <- summary(full_quasi)$coefficients[2, 2]
tstar <- qt(.975, df.residual(full_glm))
```

Interpret the quasipoisson interval in context. Which interval is wider, and which should we trust?

Inference for the mean count (λ)

To estimate the mean number of laws, λ, for a particular state:

1. Predict $\hat{\eta}$ and its standard error on the **link (log) scale**
2. Build the interval on the log scale: $\hat{\eta} \pm t^* \cdot SE(\hat{\eta})$ (use z^* for the Poisson model)
3. Exponentiate the endpoints

```
predict(full_quasi, newdata = data.frame(historical = "dem", percent_urban = 73.3),
        type = "link", se.fit = TRUE)
```

```
$fit
       1
3.057568

$se.fit
[1] 0.2312448

$residual.scale
[1] 3.349567
```

Example: Minnesota historically votes dem and has 73.3% of its population in urban areas. Find a 95% confidence interval for the mean number of laws in states like Minnesota.

Why is the interval built on the log scale rather than the count scale?

How would the interval from the standard Poisson model compare?

Testing nested models: the drop-in-deviance test

> **Likelihood ratio (drop-in-deviance) test**
>
> **Hypotheses:** H_0: _____ vs. H_a: at least one of these coefficients is not 0
> **Test statistic:** $G = D_{\text{Reduced}} - D_{\text{Full}}$, where D is the _____ of each model
> **Reference distribution:**
>
> - Poisson: $G \dot{\sim} \chi^2_d$, where d is the number of coefficients being tested
> - Quasipoisson: $F = \dfrac{G/d}{\widehat{\psi}} \sim F_{d,\ n-(p+1)}$

Example: Is `historical` needed in the model, after accounting for `percent_urban`?

```
reduced_glm <- glm(laws ~ percent_urban, data = equality_index, family = "poisson")
reduced_quasi <- glm(laws ~ percent_urban, data = equality_index, family = "quasipoisson")

anova(reduced_glm, full_glm, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: laws ~ percent_urban
Model 2: laws ~ percent_urban + historical
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        48     601.82
2        46     413.49  2   188.32 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(reduced_quasi, full_quasi, test = "F")
```

```
Analysis of Deviance Table

Model 1: laws ~ percent_urban
Model 2: laws ~ percent_urban + historical
  Resid. Df Resid. Dev Df Deviance      F    Pr(>F)
1        48     601.82
2        46     413.49  2   188.32 8.3927 0.0007811 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

State H_0, d, and your conclusion under each model. Why is the quasipoisson test less extreme?

Summary

- Poisson regression assumes the mean and variance are _____. **Overdispersion** means the variance is larger, and the Poisson SEs are too _____.
- Check with the Pearson X^2 statistic and $\widehat{\psi} = X^2/(n-(p+1))$.
- Rule of thumb: $\widehat{\psi} \approx 1$ keep Poisson; $\widehat{\psi} > 1$ use _____; $\widehat{\psi} < 1$ keep _____.
- Confidence intervals for the mean count are built on the _____ scale, then exponentiated.

R function reference

Function	Purpose
`glm(y ~ x, data, family = "quasipoisson")`	Fit a quasipoisson regression model
`sum(residuals(model, type = "pearson")^2)`	Pearson X^2 statistic
`df.residual(model)`	Residual degrees of freedom, $n-(p+1)$
`resid_xpanel(model, type = "stand.pearson")`	Standardized Pearson residuals against each predictor
`confint.default(model)`	Wald confidence intervals for the coefficients (log scale)
`predict(model, newdata, type = "link", se.fit = TRUE)`	Predicted log mean and its standard error
`anova(reduced, full, test = "LRT")`	Drop-in-deviance test for Poisson models
`anova(reduced, full, test = "F")`	Drop-in-deviance F-test for quasipoisson models